

Contact

+40 760 827 250 (Mobile)

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(LinkedIn)

Top Skills

Embedded C

Git/Gitlab

Microsoft Visual Studio Code

AUTOSAR Classic Platform

CANoe

Jira

INCA

Lauterbach TRACE32

Languages

Romanian (Native)

English (Professional Working)

German (Beginner)

Certifications

The Bits and Bytes of
Computer Networking

Responsive Web Design

Technical Support Fundamentals

Vector Certified Embedded
Associate

Robert Buzabrici-Filipescu



Senior Embedded Software Engineer at Capgemini
Engineering
Iași, Romania

Summary

Contact information :

Mobile : + 40 760 827 250

Email: filipescurobert@gmail.com

Embedded Software Engineer with experience in ANSI C. Basic knowledge about CAN communication and calibration protocol (CCP, XCP), V-cycle model, Agile methodology, Scrum, good understanding and system overview of powertrain engine vehicles.

For the next 2 years I want to focus on developing my skills with the following technologies: C++, C#, Python, Rust and whatever technology is needed for work.

So far in my career I was most productive when working in an environment that is: full of people that can and want to help you whenever you need their help, an environment full of communication between the team members.

Specialties:

Proficient: C, GIT, Gitlab

Intermediate: C++/C#, , HTML, JavaScript, CSS, MySQL,
PostgreSQL, PVCS, Canoe/Canalyzer, MKS, IMS, RTC, Trace32,
Multi, Jenkins, Linux

Basic: Java, PHP, Oracle, Doors, ESI, IMES

Experience

CAPGEMINI Engineering

Senior Embedded Software Engineer / Tech Lead Replacement

April 2022 – Present (3 years 5 months)

Iași, Romania

- Spearheaded the performance optimization and memory usage improvements in embedded modules.
- Mediated development and validation teams, improving cross-functional communication and issue resolution speed.
- Improved automated and manual test campaigns, updated CAPL scripts, and resolved issues across Mode Management and Body teams.
- Oversee embedded software functionality on bench setups using Trace32 and CANoe, contributing to successful system integration.
- Reviewed and delivered code via Git, ensuring adherence to coding standards and project timelines.
- Developed and executed unit and component tests using Google Test, improving code reliability and reducing regression issues.
- Designed and implemented missing functionalities in C/C++ using Visual Studio Code, with a focus on modularity and maintainability.
- Led the onboarding of new team members by introducing them to development processes, fostering collaboration, and providing ongoing technical guidance to ensure smooth integration and productivity.
- Assumed Acting Tech Lead responsibilities in the absence of the designed Tech Lead, ensuring project continuity, directly interfacing with client and managing project communications and delivery of technical objectives.

ENEA Software Development Services

Senior Embedded Software Engineer

April 2021 - April 2022 (1 year 1 month)

Iași, Romania

- Conducted peer code reviews and contributed to documentation for future scalability and maintainability.
- Collaborated with cross-functional teams to align software architecture with mission-critical requirements.
- Improved system responsiveness and reduced latency in command handling through efficient buffer management and interrupt-driven design.
- Utilized Xilinx SDK and worked on Zynq®-7000 SoC with dual-core ARM® Cortex™-A9 processors, optimizing performance for constrained environments.

- Engineered software modules to interface with the SpaceWire driver, enabling real-time processing of Telecommands sent from the On-Board Computer.
- Implemented a state machine to manage communication flow between the RendezVous Sensor Processing Unit and the On-Board Computer.
- Developed embedded software in C/C++ for space-grade platforms, focusing on reliability and fault tolerance.
- Prepared comprehensive technical documentation, including Interface specifications, Module descriptions, and Integration guidelines.
- Documentation efforts significantly improved cross-team alignment, facilitated smoother integration of software components, and supported successful project milestones.
- Collaborated with system architects and validation teams to ensure documentation accuracy and completeness, contributing to long-term maintainability and knowledge transfer.

RINF TECH

Software Development Consultant

June 2014 - August 2020 (6 years 3 months)

Sofia, Sofia City, Bulgaria

VISTEON Corporation

Senior Embedded Software Engineer (Contractor from SC RINF Outsourcing Solutions SRL)

June 2014 - April 2020 (5 years 11 months)

Sofia, Bulgaria

Projects : VS Daimler 2.0, MFA2 for Trucks and for Cars, MRA2 Cars (Client: Daimler) – Instrument Cluster (IC)

- Led development of embedded software in C/C++ for automotive platforms, focusing on instrument cluster functionalities.
- Utilized CANoe simulation configurations and debugger tools for testing and validation of Maintenance Interval Display (WIA = WartungsIntervallAnzeige) component.
- Designed, implemented and tested various components such as Time and Date, Odometer, Speedometer, Tachometer, Language Setting, Outside Air Temperature, Steering Wheel Switches, Coolant Temperature.
- Developed and executed Unit Tests using VectorCast, improving code reliability and reducing defects.
- Generated and integrated Franca Interface Definition Language (.fdl) files and implemented Inter-Process Communication (IPC) mechanisms to facilitate communication between Application Layer and HMI.
- Collaborated with cross-functional teams to ensure alignment with AUTOSAR RTE architecture and project specifications.

- Technologies and protocols: Autosar RTE, ANSI C, MULTI IDE, Visual C++ 6.0 IDE, PVCS, Interactive MID cluster, Inter-process communication, Renesas flashing tools for resources, Serena TeamTrack, Jenkins, Rational Team Concert (RTC), CANoe, CANalyzer

CONTINENTAL Automotive Romania SRL

Embedded Software Engineer

August 2010 - June 2014 (3 years 11 months)

- Implemented customer-specific functionalities in C/C++ based on detailed specifications and design models.
- Configured and managed diagnostic faults (DTC), freeze frames, failure reactions, and inhibition mechanisms.
- Set up and validated CAN channels and diagnostic instances, ensuring robust error management.
- Designed and executed unit tests to verify software computations and system behavior.
- Performed integration testing and documented results in quality assurance reports.
- Collaborated with function developers to resolve specification discrepancies and improve implementation accuracy.
- Worked on multiple customer projects including Ford, Renault, Volkswagen, Porsche and Audi as part of the Competence Center Iasi Team.
- Technologies and protocols: CAN communication, CCP, XCP, OSEK RTOS, V-cycle model, Powertrain systems, ISO26262.

Education

Faculty of Electronics, Telecommunications and
Information Technology, Iasi, Romania

Master's Degree, Telecommunication Networks · (2009 - 2011)

Faculty of Economy and Business Administration, "Al.I.Cuza"
University, Iasi, Romania

Bachelor's Degree, Economics and Management · (2007 - 2010)

Faculty of Automatic Control and Computer
Engineering, "Gh.Asachi" University, Iasi, Romania

Bachelor's Degree, Information Technology · (2005 - 2009)

"Mihail Kogalniceanu" Highschool, Vaslui, Romania
hs, Mathematics and Computer Science · (2001 - 2005)

"Stefan cel Mare" School no 5, Vaslui, Romania
primary school, primary school · (1993 - 2001)